

MATTHEW D. HALL

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EDUCATION

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PhD - The University of Sheffield

OCT 2017 - PRESENT, SHEFFIELD

PROJECT: Self-Reconfiguration of Modular Robots via Active Subtraction.

SUPERVISORS: Dr Roderich Gross (Natural Robotics Lab).

Dr Shuhei Miyashita (Microrobotics Lab).

AFFILIATIONS: Department of Automatic Control & Systems Engineering.

Natural Robotics Lab.

Sheffield Robotics.

SCHOLARSHIP: TUOS and EPSRC Doctoral Training Partnership National
Productivity Scholarship for course and maintenance fees.

MEng 1st Class Hons - Lancaster University

OCT 2012 - JUL 2016, LANCASTER

PROJECTS: Spatial Distribution of Blood Flow in Peripheral Vein Networks
by Non-Invasive Near-Infrared Imaging.

Design and Construction of an Optical Pickup System for an
Electric Bass Guitar.

SUPERVISORS: Dr Farid Aiouache.

Professor Graeme Burt.

PUBLICATIONS

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M. D. Hall, A. Özdemir, R. Groß, “**Self-Reconfiguration in Two-Dimensions via Active Subtraction with Modular Robots**”, in *Proc. Robotics: Science and Systems XVI (RSS)*, RSS Foundation, 2020.

A. Özdemir, M. Gauci, A. Kolling, M. D. Hall, R. Groß, “**Spatial Coverage Without Computation**”, in *Proc. 2019 International Conference on Robotics and Automation (ICRA)*, IEEE, 2019, pp. 9674–9680.

ORAL PRESENTATIONS

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2021: Departmental Postgraduate Research Symposium - **Awarded “Best Oral Presentation”** - *Virtual*.

2020: Int’l. Conf. on Robotics: Science and Systems (RSS) - *Virtual*.

2019: Self-Configuring Modular Robotics for Earth and Space - *York, UK*.

IEEE Int’l. Conf. on Robotics and Automation (ICRA) - *Montreal, Canada*.

UK & I Conf. on Transformative Robotics for Industries - *London, UK*.

Faculty Researcher Symposium - **Awarded “Best Poster Presentation”** -
Sheffield, UK.

EXPERIENCE

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Peer Reviewer for International Conferences – *Natural Robotics Laboratory*

OCT 2017 – PRESENT

Peer reviewed submitted works for a number of highly regarded international conferences, such as: Autonomous Agents and Multiagent Systems (AAMAS), Advanced Networks and Telecommunication Systems (ANTS), Distributed Autonomous Robotics Systems (DARS), Intelligent Robots and Systems (IROS), Multi-Robot Systems (MRS), Robotics: Science and Systems (RSS), Towards Autonomous Robotic Systems (TAROS).

Kept up to date with state of the art work in the field. Required the ability to disseminate information and assess the quality of written works.

Graduate Teaching Assistant – *University of Sheffield*

OCT 2017 – OCT 2020

Assisted in the planning and running of laboratory based teaching and seminar sessions for Master's and Bachelor's students.

Required an in-depth knowledge of material and the ability to clearly communicate to answer the students' questions. Organisational skills also required to effectively plan sessions for multiple groups for the semester.

Tourism Advisor – *Use-it Brussels*

JAN 2017 – AUG 2017

Part-time work within a sustainable tourism project, offering tourist information and running tours of the lesser-known parts of Brussels.

A great opportunity to thoroughly learn about the city and its culture, as well as improving my interpersonal skills.

Student Ambassador – *Lancaster University*

OCT 2014 – JUN 2016

Part-time employment assisting with university events and the regular running of tours around the university campus.

These responsibilities required excellent communication and organisational skills, and a personable yet professional demeanour.

SKILLS & INTERESTS

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PROGRAMMING: Proficient in Python, C/C++, LaTeX, Git, Shell Scripting.

Working knowledge of MATLAB, Java, HTML, CSS and VHDL.

Released a small number of customisation apps for Android.

MUSIC: Bassist and guitarist. Experience in Ableton music production.

Regular guest radio DJ for "Clear A Space" on GagaRadio.

FILM: Keen interest in film. Written, shot and edited amateur films with family and friends.

OTHER: Food and cooking, games (board or video), craft beer.

REFERENCES AVAILABLE UPON REQUEST

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